

Finite-temperature dynamic stability separates foundation machine-learning interatomic potentials that harmonic benchmarks rank equally

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Abstract

Foundation (universal) machine-learning interatomic potentials (MLIPs) are now used routinely as cheap stand-ins for DFT in high-throughput stability screening, but they are benchmarked almost exclusively against harmonic (0 K) phonons. Dynamic stability is physically a finite-temperature property, and an important class of materials, including cubic perovskites, bcc refractory metals and cubic fluorites, is harmonically unstable but thermally stabilised by anharmonicity. We test five foundation MLIPs (MACE-MP-0, CHGNet, ORB-v2, SevenNet-0, MatterSim) on this regime using a curated set of references whose finite-temperature behaviour is documented in the experimental and DFT literature, so that no new DFT ground truth is required. We introduce a cheap quantum self-consistent-harmonic (SCHA) “soft-mode free energy” screen that evaluates every imaginary commensurate mode and calls the high-symmetry phase unstable if any of them condenses, and we cross-validate it against the gold-standard multi-mode stochastic SCHA (SSCHA). We report three findings together with a practical ensemble-disagreement guardrail. (i) At the harmonic level the models divide: MatterSim and SevenNet-0 reproduce every documented soft mode (accuracy 1.00), whereas MACE-MP-0 and CHGNet soften the bcc Zr/Hf instabilities to zero (15% false-stable rate) and ORB-v2 misses the SrTiO₃ soft mode (8%). (ii) Harmonic accuracy does not predict finite-temperature accuracy in either direction: CHGNet is the worst model harmonically yet the second best at finite temperature, while MatterSim is harmonically perfect and only mid-table, and on the matched set the two are non-significantly concordant at 100 K and significantly anti-associated at 300 K. Screening every imaginary mode rather than the softest one is essential, because the deepest mode need not be the one that condenses: in SrTiO₃ the Γ ferroelectric mode is deeper than the R-point tilt yet is quantum-suppressed, and only the tilt drives the 105 K transition. (iii) Multi-mode SSCHA with an MLIP force engine is a clean gold standard for the martensitic bcc metals, but in its default deployment, with the free-energy Hessian truncated at the bubble, it systematically false-stabilises deep displacive (ferroelectric) instabilities and can diverge numerically. Theory predicts this truncation to fail for cubic metal halide perovskites; we measure the failure at benchmark scale and show it extends to oxide perovskites and, free of stochastic blow-ups, to cubic fluorites. The cheap soft-mode screen is therefore the more reliable finite-

temperature indicator for the displacive instabilities that dominate generative crystal-structure-prediction (CSP) screening.

1. Introduction

Foundation MLIPs (MACE-MP, CHGNet, ORB, SevenNet, MatterSim, M3GNet) are increasingly used as DFT surrogates in high-throughput pipelines, including the dynamical-stability filtering of generative CSP outputs. Two 2025 works define the state of the art. “Universal MLIPs are ready for phonons”¹ benchmarks MLIPs against $\sim 10^4$ DFT harmonic phonon calculations³ on largely in-distribution materials and reports a systematic potential-energy-surface (PES) softening bias, in which energy and force under-prediction soften modes and over-estimate stability. PhononBench² scores $\sim 1.1 \times 10^5$ generated structures for harmonic dynamical stability using MatterSim itself as the oracle.

Both are harmonic, that is, 0 K. Dynamic stability is a finite-temperature property, and a technologically central materials class is harmonically unstable yet thermally stabilised by anharmonicity: cubic perovskites (SrTiO_3 , BaTiO_3 , halide perovskites), bcc refractory metals ($\beta\text{-Ti/Zr/Hf}$), and cubic fluorites ($\text{ZrO}_2/\text{HfO}_2$). Harmonic phonons assign these imaginary modes, yet the high-symmetry phase is the equilibrium phase above a transition temperature, and the harmonic-only oracle that PhononBench and CSP screening rely on is what this regime breaks. Two 2025 studies probe finite-temperature reliability directly: four MACE foundation variants were benchmarked for the dynamic stability of halide double perovskites,¹⁹ and the ferroelectric-to-paraelectric transition of PbTiO_3 was used to expose a disconnect between static accuracy and dynamic reliability.²⁰ Both are confined to a single model family or a single system. What remains untested is whether the effect is architecture-general and chemistry-general: no study has classified finite-temperature dynamic stability across several independent MLIP architectures and across the distinct anharmonic families, that is displacive and antiferrodistortive oxide perovskites, halide perovskites, entropy-stabilised bcc refractory metals and cubic fluorites, under a common free-energy criterion and a temperature ladder.

Research question: do foundation MLIPs reproduce finite-temperature dynamic stability, specifically the harmonic-unstable to thermally-stabilised transition, or does PES softening make their stability calls unreliable where the harmonic approximation fails?

We pre-registered three hypotheses.

- H1 (softening to false-stable). PES softening inflates harmonic stability, so MLIPs over-report stable with a measurable false-stable rate on harmonically-unstable references. This reproduces the literature picture on our model set and anchors the validity of the harness.
- H2 (the finite-temperature gap, the contribution). Harmonic accuracy does not certify a model for finite-temperature screening, and the harmonic ranking need not carry over. (The pre-registered form was “harmonic accuracy is uncorrelated with finite-temperature accuracy”. The matched-set outcome is temperature-

dependent — non-significantly concordant at 100 K, and significantly anti-associated at 300 K — so the pre-registered form is neither cleanly confirmed nor cleanly rejected; see §3.2.)

- H3 (practical guardrail). Inter-model (ensemble) disagreement flags unreliable calls better than any single model’s self-reported energetics.

Our contributions are a finite-temperature benchmark on the anharmonic regime spanning five independent MLIP architectures and four anharmonic chemistry families; a cheap, validated free-energy screen whose criterion is the phase rather than a single mode; a cautionary result that MLIP-driven SSCHA at its default bubble truncation false-stabilises deep displacive instabilities, which confirms and extends a published theoretical prediction and changes how such cross-checks should be used; and a practical ensemble-disagreement guardrail (H3).

2. Methods

2.1 Systems and ground truth

The curated set (`configs/curated_systems.yaml`, 20 systems) comprises references whose harmonic-imaginary and finite-temperature-stable behaviour is documented in the literature, so the ground-truth label needs no new DFT:

- Oxide displacive/antiferrodistortive perovskites (4): SrTiO_3 ,¹¹ BaTiO_3 , PbTiO_3 , KNbO_3 .¹²
- Halide perovskites (3): CsPbI_3 ,¹³ CsSnBr_3 , CsSnI_3 .¹⁴
- bcc refractory metals (3): Ti, Zr, Hf (phonon-entropy stabilisation).¹⁵
- Cubic fluorites (2): ZrO_2 , HfO_2 .¹⁶
- Superionic (1): $\alpha\text{-AgI}$.¹⁷
- Quantum-paraelectric near-control (1): KTaO_3 .¹⁸
- Harmonically-stable controls (6): Si, MgO, NaCl, Cu, diamond, CeO_2 (0 K harmonic stability documented in the DFPT phonon database³).

Transition temperatures are approximate, and scoring is qualitative (the correct side of the transition, not a fitted T_c). KTaO_3 is flagged borderline (incipient ferroelectric, DFT mode ≈ 0 ; four of five MLIPs call it imaginary) and excluded from headline rates, which leaves 19 scored systems. CeO_2 and NaCl are retained as scored stable controls, since they are genuinely stable and only CHGNet marginally trips them (§3.1). The soft-mode screen and the harmonic baseline cover all 20 systems; the SSCHA grid covers the perovskites, fluorites and bcc metals. $\alpha\text{-AgI}$ is the one system where the soft-mode screen is on shaky physical ground, because a superionic with a diffusive Ag sublattice has no single frozen order parameter; its screen call is reported but flagged, and it is the natural target for a future symmetry-breaking/MD probe.

2.2 Models

MACE-MP-0,⁶ CHGNet,⁷ ORB-v2,⁸ SevenNet-0,⁹ and MatterSim¹⁰ run through a backend-agnostic ASE-calculator harness, with one virtual environment per model. Testing MatterSim directly probes PhononBench’s own oracle. ORB-v2 is a direct, float32-only model; we carry this as an explicit caveat because it manifests in both the harmonic and the finite-temperature results.

2.3 Harmonic baseline (credibility anchor)

Finite-displacement harmonic phonons (phonopy, 2×2×2 force-constant supercells) with each MLIP, classified by the minimum phonon frequency against an imaginary tolerance (default −0.1 THz). The success criterion is to reproduce the published harmonic stability split. This validates the harness and is not claimed as novel.

Metrics convention (used throughout). A call is false-stable when a model predicts the high-symmetry phase dynamically stable where the reference is unstable; this is the screening-dangerous error, because it lets an unphysical structure through a CSP filter. A call is false-unstable in the opposite case. Recall (unstable) is the fraction of genuinely-unstable reference units a method correctly flags as unstable. Rates exclude the borderline KTaO₃.

2.4 Finite-temperature soft-mode free energy (primary screen)

For each system we relax the structure, compute harmonic force constants, and enumerate **every distinct imaginary mode** on the force-constant-commensurate **q** grid by exact run_qpoints evaluation over the rational grid the supercell supports (bcc uses 6×6×6 force constants so the ω -phase $q = \frac{2}{3}\langle 111 \rangle$ is commensurate). The three acoustic branches at Γ are removed as the branches *nearest zero in magnitude*, not the three lowest: when the high-symmetry phase is unstable the soft mode lies below the acoustic zeros, so masking the lowest three would delete the very instability being sought. Modes are deduplicated over degenerate branches and symmetry-equivalent **q**, so a triply degenerate T1u triplet and the three arms of a $\langle 100 \rangle$ star each cost one E(Q) map rather than three. Each surviving mode is frozen into its minimal commensurate cell via phonopy modulation; we map the static double well E(Q) with quadratic Q sampling, fit the well-plus-barrier window, and minimise a single-mode quantum SCHA free energy over the order-parameter centroid (self-consistent Gaussian width via bracketed root finding).

Variational free energy per mode. Each frozen mode defines a one-dimensional subsystem with coordinate Q along the unit displacement pattern **u**, effective mass $M = \sum_i m_i |\mathbf{u}_i|^2$, and potential $V(Q) = aQ^2 + bQ^4 + cQ^6$ from the fit above; the Hamiltonian is $H = P^2/2M + V(Q)$. We treat it with the self-consistent harmonic approximation in its original single-mode form,^{23,24} built on the Peierls variational bound^{25,26}

$$F \leq \mathcal{F}(Q_0, \Omega; T) = F_0(\Omega, T) + \langle V \rangle_{Q_0, \sigma} - 1/2 M \Omega^2 \sigma^2,$$

where the trial state is the thermal density of a harmonic oscillator of frequency Ω displaced to centroid Q_0 , $F_0(\Omega, T) = k_B T \ln[2 \sinh(\hbar\Omega/2k_B T)]$ is its free energy, and the last term removes the trial potential counted in F_0 . The trial density is a Gaussian of mean Q_0 and quantum width

$$\sigma^2 = \frac{\hbar}{2M\Omega} \coth\left(\frac{\hbar\Omega}{2k_B T}\right),$$

which carries the nuclear quantum effects: at high T it recovers the classical $k_B T/M\Omega^2$, and at $T \rightarrow 0$ it retains the zero-point width that suppresses condensation in shallow wells (quantum paraelectricity). The Gaussian expectation of the even sextic follows from the moments $\langle Q^2 \rangle = Q_0^2 + \sigma^2$, $\langle Q^4 \rangle = Q_0^4 + 6Q_0^2\sigma^2 + 3\sigma^4$, $\langle Q^6 \rangle = Q_0^6 + 15Q_0^4\sigma^2 + 45Q_0^2\sigma^4 + 15\sigma^6$. Stationarity of $\mathcal{F}$ with respect to Ω at fixed Q_0 gives the self-consistency condition

$$M\Omega^2 = \langle V'' \rangle_{Q_0, \sigma} = 2a + 12b \langle Q^2 \rangle + 30c \langle Q^4 \rangle,$$

which we solve by bracketed root finding on σ^2 (avoiding the runaway large- σ fixed point a damped iteration can reach), and $\mathcal{F}$ is then minimised over the centroid on a Q_0 grid.

Criterion and observable. Two distinct quantities come out of $\mathcal{F}$, and we keep them separate. The **stability call** is the variational one: the mode has condensed at T if the global minimum of $\mathcal{F}(Q_0; T)$ sits at $Q_0 > 0$ — by the bound above, the displaced state then has the lower free energy. The high-symmetry structure is dynamically unstable at T if **any** screened mode condenses, and stable only when none does; dynamical stability is a property of the phase, not of one mode. The **reported frequency** is the curvature of the free energy at the symmetric point, $\omega_{\text{eff}} = \text{sign}(\mathcal{F}''(0)) \cdot [|\mathcal{F}''(0)|/M]^{\frac{1}{2}}$, the single-mode analogue of the SSCHA free-energy Hessian²¹ and a genuine signed observable ($\mathcal{F}$ is even in Q_0 , so the curvature is evaluated by a symmetric finite difference). The two can disagree, and the disagreement is physics rather than noise: for a deep double well the variational transition is first-order-like, so $\mathcal{F}(0)$ remains a *local* minimum (positive curvature) while a displaced minimum drops below it. A curvature criterion is blind to that condensation by construction — the same blindness that afflicts any free-energy Hessian evaluated at a fixed high-symmetry reference — and we record every such mode (`n_curv_blind` in the ledger) rather than folding it into either number.

Screening only a single mode — even the globally softest — conflates physically distinct instabilities. Cubic SrTiO_3 is the decisive case: its Γ ferroelectric mode is *deeper* than the R-point antiferrodistortive tilt, yet the Γ mode is quantum-suppressed and never condenses while the R tilt is what drives the 105 K transition, so a softest-mode screen inspects the wrong mode and calls the cubic phase stable. The $E(Q)$ maps are temperature-independent and cached, so each temperature is a sub-second CPU solve over all modes. A cap of 24 modes per unit bounds the cost; it binds on 20 of the 400 units, every one of which is already called unstable with at least three condensing modes, so it cannot affect any call (a cap can only ever create a *false-stable*, which would require all 24 screened modes to be non-condensing). Three earlier finite-temperature routes (hand-

rolled TDEP,⁵ one-shot hiPhive, and rattled-MD) were implemented and discarded after they failed the SrTiO₃ gate; see the ESI.

Approximations, declared. Table 1 states what the screen neglects, the expected direction of the bias, and where the consequence is visible in our own data.

Table 1 The screen's approximation ledger.

#	Approximation	What it excludes	Expected bias	Where it shows
A1	One mode at a time	Mode–mode coupling and cooperative condensation	Either sign; absolute T* unreliable	PbTiO ₃ T* ordering failure (§3.2)
A2	Commensurate q only	Instabilities at incommensurate or finer-grid q	False-stable if the true soft mode is missed	R point needs even cells (§3.5); bcc ω needs 6×6×6 FCs
A3	Sextic fit on a well+barrier window	Steep-wall anharmonicity beyond Q ⁶	Barrier-shape error at large Q	Fit window sensitivity (ESI)
A4	Gaussian (SCHA) trial state	Tunnelling, non-Gaussian density; renders deep-well transitions first-order-like	Curvature misses condensation	n_curv_blind rows; §3.3 mechanism
A5	Static E(Q) in a clamped cell	Thermal expansion and strain coupling	Under-stabilises martensitic systems	bcc labelled via SSCHA instead (§3.3)
A6	Mode cap (24)	Modes beyond the 24 most imaginary	Could only create false-stables	Binds on 20/400 units, all already unstable (§2.4)

Validation gate. Cubic SrTiO₃ resolves three distinct imaginary commensurate modes, and the screen reports which of them condenses. At 100 K the Γ ferroelectric mode does **not** condense ($Q_0 = 0$), which is the physically correct quantum-paraelectric result, while the R = (½,½,½) tilt does ($Q_0 = 0.14 \text{ \AA}$); by 300 K neither condenses. The phase is therefore called unstable at 100 K and stable at 300 K, bracketing the experimental transition at 105 K. The gate also exercises the criterion/observable distinction above: at 100 K the condensing tilt still has a *positive* symmetric-point curvature (+0.92 THz, hardening to +1.43 THz at 300 K),

so this is a first-order-like condensation that the free energy’s argmin detects and its curvature does not — the same mechanism, inside the cheap screen, that defeats the fixed-reference SSCHA Hessian in §3.3. Three of the five models (MACE-MP-0, MatterSim, SevenNet-0) reproduce the R-tilt condensation; CHGNet never condenses the tilt and ORB-v2 does not select the R point at all. Because the gate identifies *which* instability each model captures rather than returning a single number, it is a per-model test rather than a single-model demonstration.

2.5 Multi-mode SSCHA (gold-standard cross-check)

The full stochastic SCHA (python-sscha + cellconstructor)⁴ uses the MLIP as force engine: an ASE finite-displacement harmonic dynamical matrix, then ForcePositiveDefinite, then stochastic SCHA relaxation of the auxiliary dynamical matrix at T (root2 representation, with a per-population step cap to force ensemble regeneration on noisy large cells), then a dedicated ensemble at the converged matrix, and finally the free-energy (physical) Hessian. The minimum Hessian frequency excluding the three acoustic modes is the anharmonic dynamic-stability indicator. Unlike the free-energy screen, which treats each imaginary mode independently, this captures the coupled multi-mode phonon entropy and is reliable for entropy-stabilised martensitic transitions.

3. Results

3.1 Harmonic baseline: the models divide (H1)

The five models divide on the harmonic set (19 scored systems, KTaO₃ excluded):

Model	Accuracy	False-stable rate	False-unstable rate
MatterSim	1.000	0.000	0.000
SevenNet-0	1.000	0.000	0.000
MACE-MP-0	0.895	0.154	0.000
ORB-v2	0.895	0.077	0.167
CHGNet	0.789	0.154	0.333

MatterSim and SevenNet-0 reproduce every documented soft mode with large imaginary frequencies. MACE-MP-0 and CHGNet each carry two false-stable calls (rate 0.154) on the bcc Zr and Hf soft modes, softened to -0.00 THz; this is the PES-softening bias of the literature, localised to specific instabilities. ORB-v2 carries one false-stable (rate 0.077): it correctly flags bcc-Zr (-0.43 THz) and bcc-Hf (-0.19 THz) unstable but reads SrTiO₃ at $+0.00$ THz. ORB-v2 (a direct-force model, run in its default float32 precision) softens in both directions; besides that false-stable it falsely calls MgO unstable (-1.07 THz), its one false-unstable on the oxide side. CHGNet carries the only non-ORB false-unstables (rate 0.333), CeO₂ (-0.27 THz) and NaCl (-0.24 THz), both genuinely stable controls tripped just past the -0.1 THz tolerance; both flip back to stable at a tolerance of ≈ 0.25 THz, so they are finite-displacement noise rather than a real instability.

Because every binary call depends on the imaginary tolerance, we sweep it (Fig. 1). A strict tolerance (0 THz) floods false-unstables (29 calls) as near- Γ finite-displacement noise dominates, while a loose tolerance (0.3 THz) inflates false-stables (7 calls); the default -0.1 THz sits in the stable basin between them (5 false-stable, 3 false-unstable), and the split holds across 0.05–0.2 THz. The useful point is that the softening toward zero is the physics: per-system minimum frequencies, not binary rates, are the right reporting unit (Fig. 2). This reproduces the published picture and validates the harness.

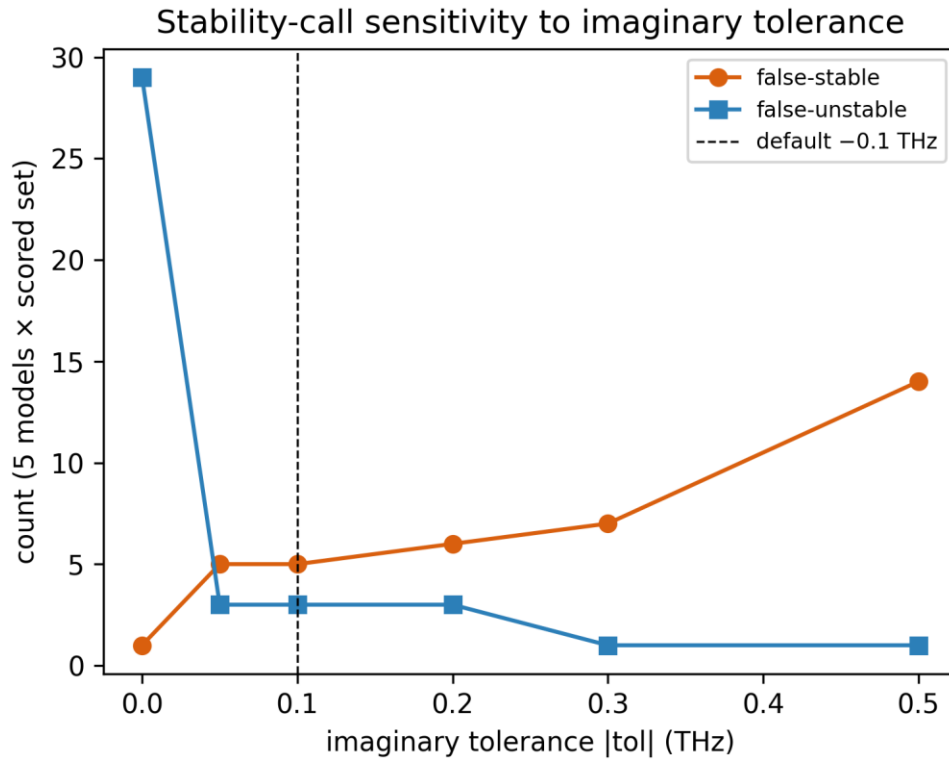


Fig. 1 Harmonic false-stable and false-unstable call counts versus the imaginary-frequency tolerance (§3.1). The default -0.1 THz sits in the stable basin between the false-unstable flood at strict tolerance and the false-stable inflation at loose tolerance.

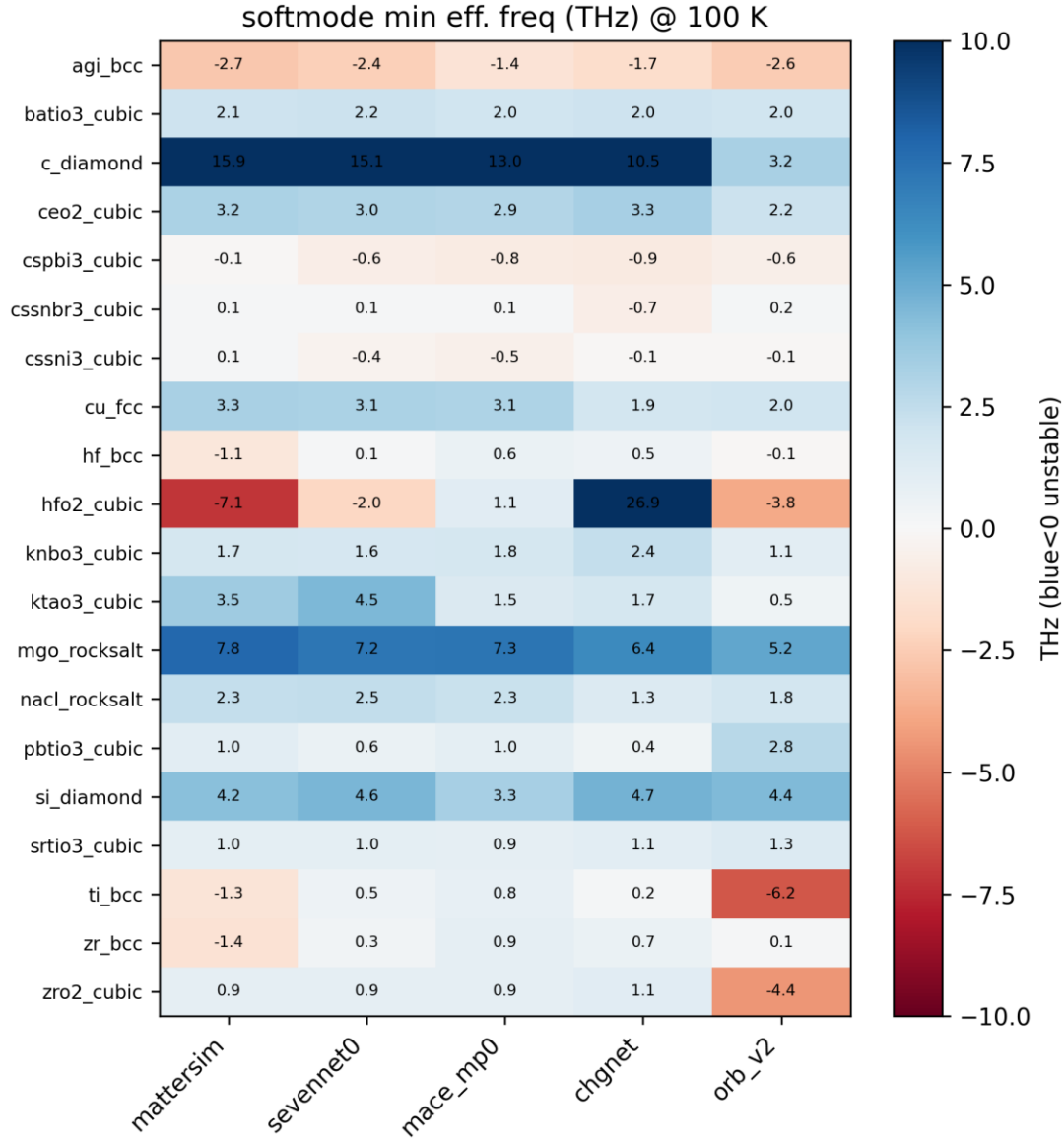


Fig. 2 Per-system minimum effective frequency across the five models (§3.1–3.2). The softening toward zero of the harmonically-unstable systems is the physics; per-system minima, not binary rates, are the right reporting unit.

3.2 Finite-temperature soft-mode screen (H2)

On the ferroelectric perovskites at $T \leq 300$ K, far below every transition temperature, the cubic phase is definitively dynamically unstable, and the multi-mode screen correctly calls it unstable in 16 of 30 model units (recall 0.53). The screen recovers displacive instability that harmonic accuracy does not predict and that, as §3.3 shows, the gold-standard SSCHA misses more often still. Across the other anharmonic families the screen is stronger: it reaches recall 0.92 on the halide perovskites and 1.00 on the cubic fluorites, and on the antiferrodistortive SrTiO_3 tilt it reaches 0.60. On the six harmonically-stable

controls it is correct on all 120 model units, and finds **zero** imaginary commensurate modes in every case, so it never manufactures an instability.

Multi-anchor comparison against experiment. Beyond the SrTiO₃ gate (§2.4) we compare the screen’s predicted stabilisation temperature T^* (the lowest ladder T at which the cubic phase is called stable) with the experimental transition temperatures. The comparison is only partly successful and we report it as such. SrTiO₃ ($T_c \approx 105$ K) stabilises earliest, at $T^* = 100\text{--}300$ K, and BaTiO₃ (393 K) and KNbO₃ (708 K) follow at $T^* = 300$ K for four of five models (ORB-v2 at 600 K), so the SrTiO₃-below-ferroelectric ordering holds. PbTiO₃ (763 K), which has the highest transition temperature of the four, does **not** follow: T^* is 100 K for MACE-MP-0, MatterSim and ORB-v2, 600 K for SevenNet-0 and 900 K for CHGNet. A single-mode treatment is not expected to reproduce an absolute T_c , but the PbTiO₃ failure is a genuine ordering failure rather than a scale error, and it is the clearest limitation of the screen in this work. T^* is therefore reported as a diagnostic (full table S3), and the SrTiO₃ gate together with the control performance — not the T^* ordering — is what licenses the screen as the reference in §3.3.

Per-model false-stable rates on the displacive/anharmonic set (non-bcc, non-borderline, $T \leq 300$ K; bcc excluded because its thermodynamic- T_c label is the wrong reference for dynamic stability, §3.3):

Model	Finite-T false-stable rate	Finite-T accuracy	(Harmonic accuracy)
SevenNet-0	0.125	0.900	1.000
CHGNet	0.188	0.867	0.789
MatterSim	0.250	0.833	1.000
MACE-MP-0	0.250	0.833	0.895
ORB-v2	0.312	0.800	0.895

Read against the harmonic ranking (final column), the ordering is not preserved but neither is it inverted, and we are explicit about this because a naive comparison invites a cleaner story than the data support. The matched non-bcc, non-borderline comparison ($n = 75$ system×model pairs) is temperature-dependent. At $T = 100$ K the two are weakly and non-significantly concordant: $\phi = 0.149$ with McNemar exact $p = 0.73$, and a matched per-model accuracy rank correlation of $\rho = +0.65$. At $T = 300$ K the relationship reverses and becomes significant: $\phi = -0.129$ with McNemar exact $p = 0.007$, driven by 17 harmonically-correct units that are mis-called at finite temperature against only 4 the other way. Harmonic correctness therefore does not transfer, and at the higher temperature it is significantly *anti*-associated with finite-temperature correctness on this set.

The defensible H2 statement is that harmonic accuracy does not predict finite-temperature accuracy, in either direction. The clearest single demonstration is CHGNet: it is the **worst** model harmonically (0.789) yet the second **best** at finite temperature (0.867), while MatterSim is harmonically perfect (1.000) and only mid-table at finite temperature

(0.833). A top harmonic score on the npj/PhononBench benchmarks therefore does not identify the best finite-temperature screener, and a poor one does not disqualify a model. We note explicitly that SevenNet-0 leads *both* layers, so the relationship is not a simple inversion and we do not claim one; an earlier version of this analysis, computed before the mode-selection correction described in §2.4, reported that the harmonic leaders sat behind MACE-MP-0 and CHGNet, and that specific ordering does not survive the correction. ORB-v2 remains the weakest finite-temperature screener (0.800) and the only model that fails to select the correct instability in SrTiO₃ (§2.4), consistent with its float32 direct architecture over-softening the PES, though our design cannot separate the architecture from the precision. The strongest evidence that finite temperature is a distinct, harder regime comes not from this ranking comparison but from the per-family recall above and the SSCHA failure in §3.3.

3.3 SSCHA: clean for bcc, false-stable for perovskites at the default truncation (the cautionary result)

We ran multi-mode SSCHA across both families to cross-validate the screen. The result is asymmetric and is itself a central finding. We state at the outset that this concerns SSCHA as it is realistically deployed with an MLIP force engine for a fixed high-symmetry reference, namely ForcePositiveDefinite initialisation, a 2×2×2 cell, the v4=False free-energy Hessian, and automatic stochastic relaxation, which is the recipe a practitioner escalating from a cheap screen would actually run. It is not a claim that the SSCHA formalism is wrong; the formalism anticipates this failure, and the expensive v4 Hessian is its own prescribed remedy, but that remedy is impractical at screening scale (see the root cause below). The finding is that the default escalation path, at the truncation its implementation ships with, is unsafe in this regime.

bcc metals: SSCHA is a clean gold standard, and the screen agrees with it where it counts. Across Ti/Zr/Hf × 5 models × 5 temperatures (75 runs) SSCHA produced zero numerical failures, with minimum free-energy-Hessian frequencies in a physical [0.06, 2.1] THz range. All five models anharmonically stabilise the bcc phase by ≤50 K (the dynamic-stabilisation temperature, which is far below the thermodynamic transition because bcc-to-hcp/ω is a martensitic, strain-coupled first-order transition; using the thermodynamic T_c to label dynamic stability is the wrong comparison and is what makes the models look false-stable on bcc). The margin to the stability boundary discriminates the models: ORB-v2 hugs it (0.06–0.28 THz on Ti/Hf), SevenNet-0's Hf sits at ~0.3 THz, while MatterSim and MACE-MP-0 are firmly stable (1.2–2.1 THz) (Fig. 3). Against the screen, the two methods agree on the stability call in 35 of 45 paired bcc units (sign agreement 0.78; 30 of 36, 0.83, excluding ORB-v2), while the magnitudes are not rank-correlated (Spearman ρ = 0.11): a single-mode symmetric-point curvature and a multi-mode free-energy Hessian are different observables away from the sign change, so the cross-validation statistic we report is the call agreement, not a magnitude correlation (Fig. 4).

Ferroelectric perovskites: SSCHA systematically false-stabilises, and its deepest wells do not run at all. On the same FE perovskites at T ≤ 300 K where the screen reaches 0.53 recall,

SSCHA correctly identifies the instability in only 5 of the 27 units that return a physical number (recall 0.19). Of the 30 attempted units, one blows up numerically and two fail outright; across the full FE grid seven units die at cellconstructor symmetry or ensemble assertions, six of them on PbTiO_3 — the deepest wells in the set (ORB-v2 at every temperature, MatterSim at 600 and 900 K). In the earlier, unpinned measurement these same units produced silent numerical blow-ups to -2×10^6 THz; in the pinned re-measurement they fail loudly instead, which is the better behaviour but the same verdict: the method does not return usable answers where the instability is deepest, and precision alone does not account for it (the failures span float32 and float64 models). This is the cautionary result: the expensive gold standard is less reliable than the cheap screen in the displacive regime that dominates generative-CSP outputs — recall 0.19 against 0.53, with the deepest-well units unable to complete (Fig. 5).

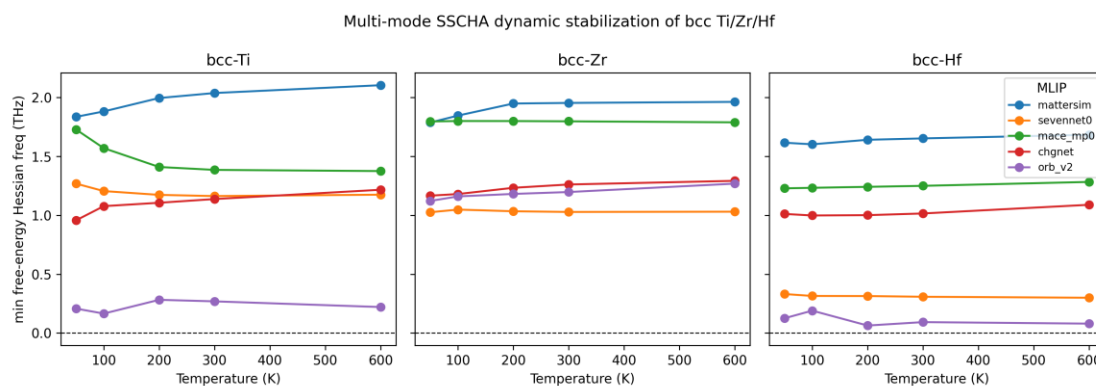


Fig. 3 Multi-mode SSCHA dynamic-stabilisation curves for bcc Ti/Zr/Hf, five models, versus temperature (§3.3). All models stabilise the bcc phase by ≤ 50 K; the margin to the stability boundary (MatterSim/ORB-v2 hugging ~ 0.4 THz, MACE-MP-0 firmly stable ~ 1.8 THz) discriminates the models.

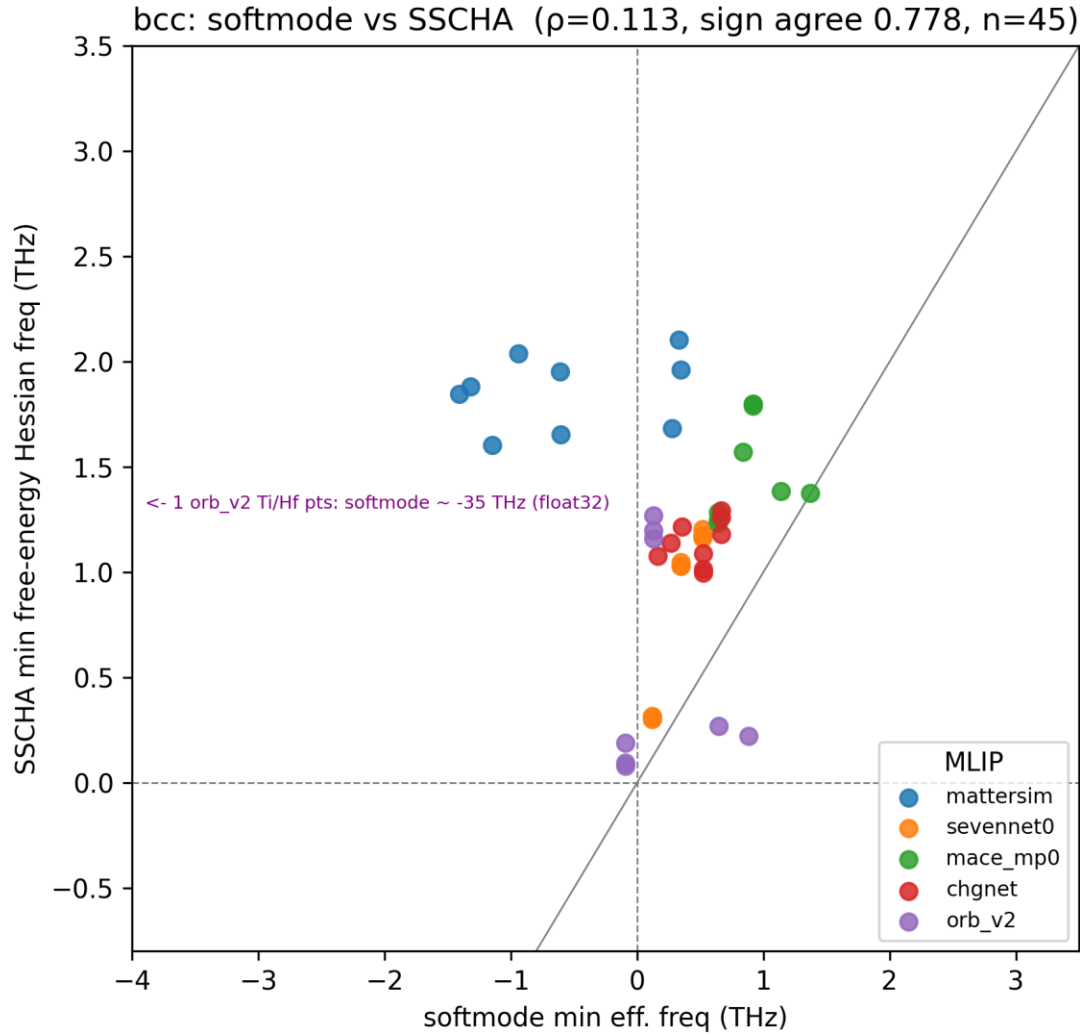


Fig. 4 Soft-mode screen vs gold-standard SSCHA minimum frequency on bcc (§3.3): the cheap screen tracks SSCHA on the family where the gold standard is trustworthy.

Root cause: the fourth-order truncation, not the reference structure. A controlled diagnostic on cubic BaTiO_3 (MACE-MP-0, 100 K) isolates the mechanism, and the SCHA literature identifies it precisely. The auxiliary SCHA matrix Φ is positive-definite by construction, since a normalisable Gaussian trial state requires it, so it can never soften and searching its eigenvalues for an instability is meaningless.^{21,22} The object that *can* detect a displacive transition from a fixed high-symmetry reference is the free-energy Hessian, $\partial^2 F / \partial \mathbf{R} \partial \mathbf{R} = \Phi + \Phi^{(3)} \Lambda [1 - \Phi^{(4)} \Lambda]^{-1} \Phi^{(3)}$, which was derived for exactly that purpose and demonstrated on the ferroelectric transitions of SnTe and GeTe.²¹ Our diagnostic shows the failure enters at the truncation: the harmonic soft mode is -5.6 THz (correctly unstable), yet with the fourth-order term dropped, which is the bubble approximation and the python-sscha default, the Hessian returns $+2.87$ THz and the call is false-stable. That is the regime in which the bubble is known to fail, since it breaks down for the cubic phase of metal halide perovskites and produces qualitatively incorrect results there, and $\Phi^{(4)}$

plays a significant role around the transition so that the complete RPA-like resummation is needed to determine phase stability.²² Our contribution is to measure that breakdown at benchmark scale and to show it is not confined to halides: it holds for oxide perovskites, and for cubic fluorites it holds in a numerically clean form with zero stochastic blow-ups (above), which separates a methodological limit from a numerical one. Enabling the fourth-order term (`include_v4=True`) is therefore the physically correct escalation, consistent with that RPA requirement, but it ran for more than 18 min on a single unit without finishing (tens of hours per unit at grid scale) and is numerically viable only in float64, which is the practical barrier at screening scale. We therefore use SSCHA as the bcc gold standard and the experimental transition temperature (the SrTiO₃ gate, §2.4) as the validation for the perovskite screen. We add one structural observation from our own data: even within the single-mode screen, deep wells condense first-order-like — the free energy develops a lower displaced minimum while its curvature at the symmetric point stays positive (§2.4, `n_curv_blind`) — and a fixed-reference Hessian criterion of any order is blind to a condensation of that character. The screen survives it because it minimises over the centroid rather than reading a curvature; a fixed-reference SSCHA deployment has no analogous escape short of relaxing the centroids, which changes the question being asked.

Cubic fluorites confirm the failure is systematic, not numerical. Cubic ZrO₂ (the >2600 K phase, harmonically unstable via the X-point oxygen mode at all temperatures studied) is a cleaner test because its instability is shallower than the FE perovskites and does not trigger the float32 blow-ups. The soft-mode screen calls both fluorites unstable at every temperature and for every model (recall 1.00), correctly and consistently. SSCHA instead returns +2.0 to +3.3 THz at 100 K for all five models on ZrO₂ (false-stable) and then destabilises with temperature for three of them (MACE-MP-0 -2.7 THz, SevenNet-0 -1.3 THz and ORB-v2 -24.8 THz at 900 K), which is both the wrong sign at low T and the wrong temperature trend; CHGNet and MatterSim instead remain positive at 900 K (+1.2 and +2.3 THz), so they are false-stable across the whole ladder. Because SSCHA fails the same way on a numerically well-behaved, shallower instability, the false-stable behaviour is intrinsic to the truncated fixed-reference deployment (see the root cause above) rather than an artifact of the deepest perovskite wells. HfO₂ shows the same pattern (all models +1.9 to +2.6 THz at 100 K, then destabilising to -4.6/-10.0 THz for MACE-MP-0/ORB-v2 by 900 K). Across both fluorites SSCHA produced zero numerical blow-ups and zero failed units, yet the same systematic false-stable, which confirms that the failure is methodological, not numerical.

3.4 Ensemble disagreement as a guardrail (H3)

Because no single model is reliable across the set, we test whether cross-model disagreement flags the units where the majority-vote consensus finite-temperature call is wrong. On the non-bcc, non-borderline soft-mode units ($n = 60$; bcc excluded because its thermodynamic-T_c label is the wrong reference for dynamic stability, §3.3), the majority-vote consensus is wrong on 20.0% of units. The disagreement signal separates these: on the 14 units where the five models split on the stable/unstable call, the consensus error

rate is 0.57, against 0.09 on the 46 unanimous units, a 6.6× enrichment. As a ranked predictor of consensus error, the binary stable/unstable vote split reaches AUC 0.76, whereas the continuous cross-model frequency standard deviation carries no usable signal (AUC 0.36, below chance on this set; Fig. 6).

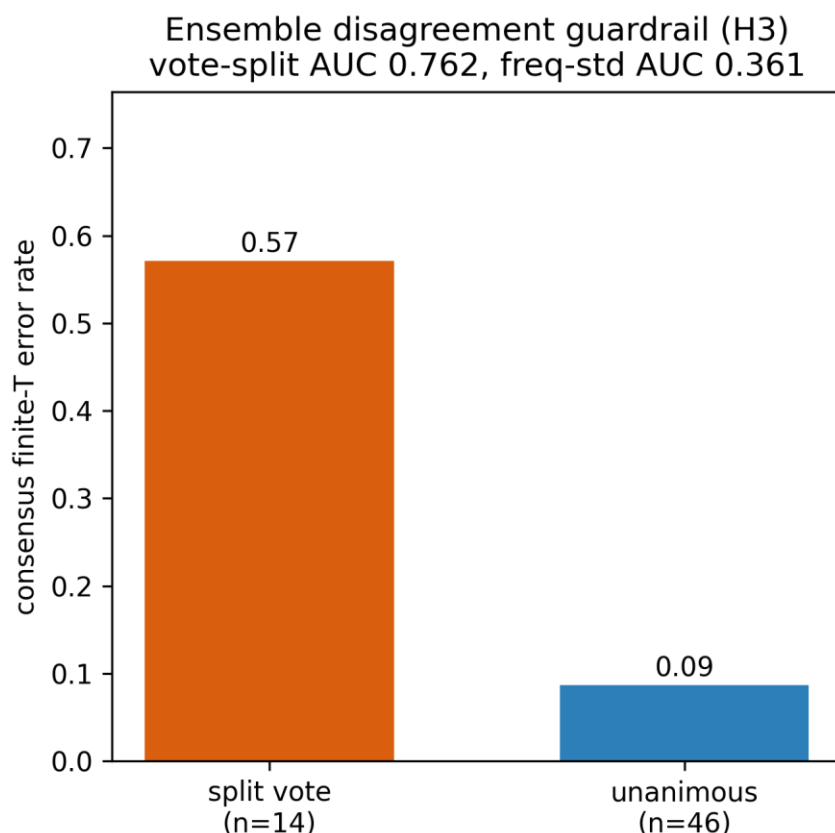


Fig. 6 Ensemble-disagreement guardrail (§3.4): the discrete inter-model vote split predicts consensus error (AUC 0.76) while the continuous cross-model frequency spread does not (AUC 0.36, below chance).

This refines H3 into a rule with a caveat: the discrete inter-model vote split is a useful, cheap guardrail (flag any candidate on which the foundation-MLIP ensemble disagrees), but the continuous frequency spread that one might naively threshold is not. The physical reading is that disagreement concentrates near the stability boundary, where split votes coincide with frequencies straddling zero and the call is both most uncertain and most error-prone.

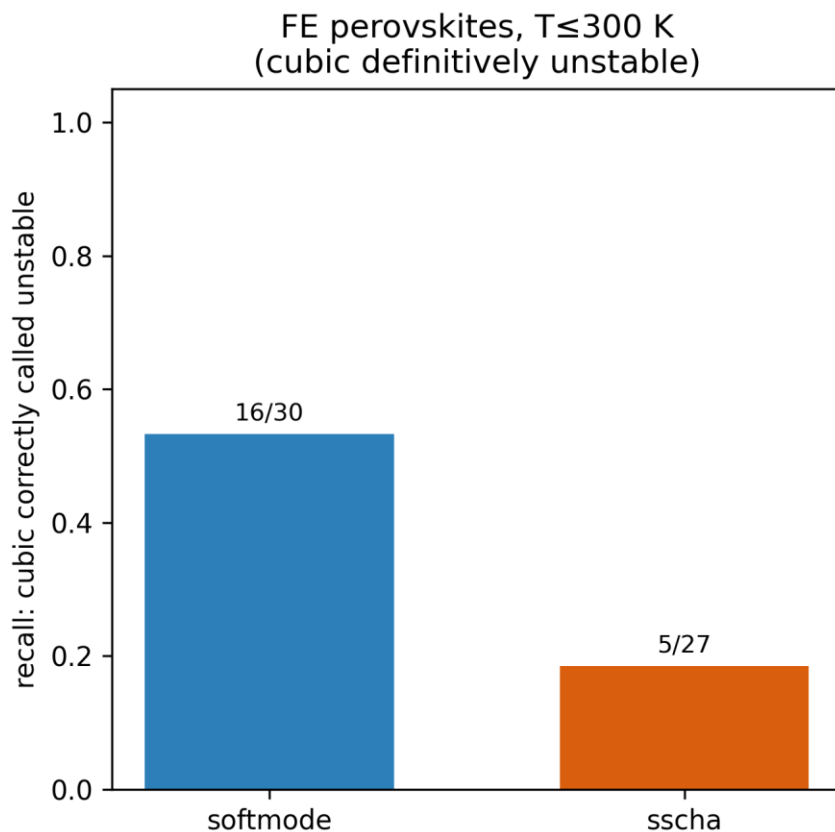


Fig. 5 Recall of the displacive (ferroelectric-perovskite) instability at $T \leq 300$ K: the cheap multi-mode soft-mode screen (0.53) versus the expensive SSCHA (0.19) (§3.3). The cautionary result is that the gold standard is less reliable than the screen in the regime that matters most.

3.5 Robustness: stochastic noise and finite size

Stochastic reproducibility. Repeating the bcc-Zr / MACE-MP-0 / 100 K SSCHA with four independent random seeds gives a minimum free-energy-Hessian frequency of $+1.798 \pm 0.001$ THz (range $[+1.797, +1.800]$). The stochastic noise (≈ 0.001 THz) is two-to-three orders of magnitude smaller than the cross-model bcc margins (§3.3), so those margins are real signal rather than sampling noise. The same unit re-measured in the pinned environment, with the SSCHA initialiser rebuilt from phonopy force constants, returns $+1.80$ THz at every temperature on the ladder, so the number also survives an independent environment and initialiser.

Finite size. The SSCHA stability call is insensitive to supercell size on the cases where the test is well-posed (results/convergence_study.parquet): for bcc-Zr the dynamic-stabilisation verdict holds from $2 \times 2 \times 2$ to $3 \times 3 \times 3$ ($+1.80$ to $+1.56$ THz at 100 K, $+1.80$ to $+1.55$ at 300 K). We report the soft-mode side of that study as superseded rather than as evidence: those runs were produced by the earlier single-mode selection, before the

correction described in §2.4, and have not been repeated at $3\times 3\times 3$. One caveat is intrinsic to the physics rather than the method, and it is now the central reason the multi-mode criterion is necessary: the SrTiO_3 antiferrodistortive instability lives at the zone-boundary R point = $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, which is commensurate only with even supercells, so a $3\times 3\times 3$ cell is blind to it by construction. A $2\times 2\times 2$ -versus- $3\times 3\times 3$ comparison is therefore not a valid convergence test for R-point systems, and the definitive even-cell test ($4\times 4\times 4$, a ~ 320 -atom SSCHA) is left to future work. The §3.3 SSCHA false-stable is not a missing-q artifact: it occurs for the Γ ferroelectric mode of BaTiO_3 , which is present in every cell including the $2\times 2\times 2$ used, so the failure is the fourth-order truncation identified in the root-cause diagnostic, not finite size.

4. Discussion

The harmonic layer (§3.1) reproduces the literature and confirms H1: PES softening produces a real false-stable rate, localised to specific instabilities (bcc Zr/Hf) rather than uniform. The finite-temperature results support H2, though with a temperature-dependent statistic rather than a single clean effect. Harmonic accuracy is non-predictive of finite-temperature accuracy, and the standard escalation (“if in doubt, run SSCHA”) fails on the displacive regime, because MLIP-driven SSCHA at its default bubble truncation false-stabilises deep double wells. The practical recommendation inverts the usual cost/accuracy intuition: for screening harmonically-unstable displacive candidates, the cheap all-imaginary-mode free-energy screen is the more reliable indicator, and a default-truncation SSCHA escalation should be reserved for martensitic and entropy-stabilised cases, or else run with the fourth-order resummation its own theory prescribes.

Limitations. Ground-truth transition temperatures are approximate and scoring is qualitative. SSCHA cells are $2\times 2\times 2$ (finite size), so dynamic-stabilisation temperatures are approximate, though the cross-model comparison at fixed cell is valid. ORB-v2’s float32-only direct architecture is an outlier in both layers and is flagged throughout rather than excluded. The screen is a single-mode treatment applied mode-by-mode: it captures each instability independently but not their coupling, which is the most likely reason the predicted stabilisation temperature fails to order PbTiO_3 correctly (§3.2). Both the screen (0.53) and SSCHA (0.19) miss a substantial fraction of the ferroelectric-oxide instabilities, so the comparison identifies the better of two imperfect methods rather than a solved problem. Finally, $\alpha\text{-AgI}$ is modelled as an ordered CsCl-type approximant of the bcc iodine sublattice; that is not the disordered superionic α phase, and its screen call should be read as a placeholder.

5. Conclusions

Finite-temperature dynamic stability separates foundation MLIPs that the harmonic benchmarks used to certify them rank equally. A cheap quantum free-energy screen, applied to every imaginary commensurate mode rather than to the softest one, recovers finite-temperature behaviour that harmonic accuracy does not predict, and it outperforms MLIP-driven SSCHA at its default bubble truncation on the deep displacive instabilities that

dominate generative-CSP screening. Which mode is examined matters as much as which method is used: the deepest instability need not be the one that condenses, and in SrTiO_3 it is not. Harmonic accuracy does not certify a model for finite-temperature use, and neither does an unexamined SSCHA cross-check.

Data availability

The code supporting this article, together with the per-unit results ledger, is openly available in the repository at <https://github.com/fronkt/mlip-dynamic-stability> and archived at Zenodo at <https://doi.org/10.5281/zenodo.20805799> (concept DOI, resolving to the latest version). The production results regenerate from `results/ledger.parquet` (per-unit hashed, resumable) and the finite-size runs from `results/convergence_study.parquet`; figures via `scripts/make_figures.py`, analysis in `mlip_dynstab/analysis.py`, the SSCHA root-cause diagnostic in `scripts/sscha_v4_diag.py`, and the stochastic-reproducibility study (§3.5) via `scripts/sscha_repro.py` (its per-seed frequencies print to the run log rather than to the ledger). The SSCHA results reported here are a full re-measurement (`scripts/run_sscha_v2.py`, method version 3) of the original 208-unit grid in the pinned environments, after two defects were found in the original analysis path: an acoustic-mode identification inversion (the three translational modes were selected by lowest frequency rather than smallest magnitude, which for an unstable phase discards the soft mode itself), and an unrecorded software environment. The superseded generations are retained in the append-only ledger — `mlip_dynstab.analysis.canonical` selects the current one — together with the in-place derived correction (`scripts/fix_sscha_acoustic.py`, *_v1 columns) and the uncorrected snapshot (`results/ledger.parquet.pre-d1-fix`), so every stage of the correction is auditable.

The finite-temperature screen results reported here are the multi-mode grid described in §2.4 (5 models × 20 systems × 4 temperatures = 400 units). The ledger is append-only and also retains the superseded single-mode rows; `mlip_dynstab.analysis.canonical` selects the current generation, and every figure and statistic in this article is computed through it. The unit hash includes the method version and the mode cap, so a re-run under a changed algorithm records new rows rather than silently reusing old ones. Each model was run in its own pinned environment, with `pip freeze` manifests deposited as `envs/lock-<model>-2026-08-16.txt`; the 220 cached E(Q) maps are included so the temperature solves regenerate without a GPU.

Author contributions

F.C. conceived the study, implemented the benchmark harness and analysis, performed all computations, and wrote the manuscript.

Conflicts of interest

There are no conflicts of interest to declare.

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